Guided Notes — Moving Objects Have Energy

Chapter 2 · Grade 4 Energy Transfer and Transformation

Name: _____ Date: _____

BIG QUESTION: How are energy, change, and movement of objects related?

WHAT WE OBSERVED

In our marble demo/investigation, write what happened.

CORE IDEA

A moving object has energy of motion.

MOTION

Motion is the process of an object changing position.

In my own words: motion means…

ENERGY OF MOTION

Energy of motion is the energy an object has while it is moving.

Our Moving Marbles evidence: the target marble changed position, so the moving marble had energy of motion.

THE MODEL

MOVING OBJECT → CAUSES CHANGE → EVIDENCE OF ENERGY

In my own words:

BEFORE / AFTER DIAGRAM

Draw the marble before and after it hits the target. Use arrows to show motion.

BEFORE	AFTER

OBSERVATION VS INFERENCE

Observation (what I saw)	Inference (what I think)
The target marble moved.	The moving marble had energy.

EXAMPLE / NON-EXAMPLE

Shows energy of motion (example)	Does NOT (non-example)

LAB CONNECTION

Moving Marbles — one thing I observed and the evidence:

CUMULATIVE CONNECTION

Energy can cause change. (From Chapter 1) A moving object is one thing that has energy. (Chapter 2)

EXIT REFLECTION

Today I learned…
